

AMAZONTECH - Gerencia de Infraestrutura e Seguranca

RELATORIO TECNICO - Laboratorio 04B

Manutencao Continua da Politica de Hardening - WEB-01

Equipe: Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

Curso: Windows e Linux Hardening | Data: 06/08/2026 | Lynis HI=69

1. Objetivo da Atividade

Avaliar o estado atual do WEB-01 em operacao assistida e estabelecer rotinas de manutencao continua: atualizacao do SO, gerenciamento de vulnerabilidades, auditoria de configuracao e monitoramento de integridade. Hardening nao termina na implantacao; sem ciclo continuo, patches atrasados e desvios de config reabrem a superficie de ataque.

2. Caracterizacao do Ambiente

Distribuicao Linux	Debian GNU/Linux 13 (trixie)
Versao	13.6
Maquina / hostname	WEB-01 / MachadoPC
Ambiente	Laboratorio de Hardening (ciclo Labs 01-04A concluidos)
Data da atividade	06/08/2026
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

3. Atualizacao do Sistema Operacional

Ferramenta	Finalidade	Resultado obtido
apt update	Sincronizar indices	Hit em trixie, trixie-updates, trixie-security, backports
apt list --upgradable	Listar pendencias	1 pacote: libexpat1 (stable-security)
apt upgrade	Aplicar atualizacoes	libexpat1 2.7.1-2 -> 2.8.2-1~deb13u1 (security)

Quantidade de atualizacoes: 1. Relacionadas a seguranca? [X] Sim — origem Debian-Security. Repositorios acessados sem erro. Prioridade: security antes de updates funcionais. Em producao: janela, rollback e testes; neste lab a atualizacao foi aplicada (baixo impacto).

Terminal - Evidencia 2 - apt update

```
root@MachadoPC:~# apt-get update
Hit:1 http://ftp.debian.org/debian trixie-backports InRelease
Hit:2 http://deb.debian.org/debian trixie InRelease
Hit:3 http://security.debian.org/debian-security trixie-security InRelease
Hit:4 http://deb.debian.org/debian trixie-updates InRelease
Reading package lists...
```

Figura - Evidencia 2 - apt update

```
Terminal - Evidencia 3 - apt list --upgradable

root@MachadoPC:~# apt list --upgradable 2>/dev/null
Listing...
libxpat1/stable-security 2.8.2-1~deb13u1 amd64 [upgradable from: 2.7.1-2]

Total upgradable (excl header): 1
```

Figura - Evidencia 3 - apt list --upgradable

```
Terminal - Evidencia 4 - apt upgrade (security libxpat1)

root@MachadoPC:~# apt-get -s upgrade 2>&1 | head -60
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages will be upgraded:
  libxpat1
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Inst libxpat1 [2.7.1-2] (2.8.2-1~deb13u1 Debian-Security:13/stable-security [amd64])
Conf libxpat1 (2.8.2-1~deb13u1 Debian-Security:13/stable-security [amd64])

root@MachadoPC:~# # Decisao: aplicar atualizacoes de seguranca no lab
root@MachadoPC:~# apt-get -y upgrade 2>&1 | tail -40
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages will be upgraded:
  libxpat1
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 122 kB of archives.
After this operation, 23.6 kB of additional disk space will be used.
Get:1 http://security.debian.org/debian-security trixie-security/main amd64 libxpat1 amd64 2.8.2-1~deb13u1 [122 kB]
Fetched 122 kB in 0s (462 kB/s)
(Reading database ...
(Reading database ... 5%
(Reading database ... 10%
(Reading database ... 15%
(Reading database ... 20%
(Reading database ... 25%
(Reading database ... 30%
(Reading database ... 35%
(Reading database ... 40%
... (saida truncada) ...
```

Figura - Evidencia 4 - apt upgrade (security libxpat1)

4. Gerenciamento de Vulnerabilidades

Ferramenta	Resultado observado	
ubuntu-security-status	N/A no Debian; equivalente: 547 pacotes ii + repo trixie-security ativo	
debsecan	381 CVEs unicos / 815 linhas; inclui openssh-server/client e libc6	
dnf updateinfo	N/A (Rocky); no Debian usamos security pocket + debsecan	
Vulnerabilidade / pacote	Criticidade	Acao recomendada
libxpat1 (security update)	Alta (security)	Aplicada neste ciclo
openssh-server/client (CVEs debsecan)	Alta (servico exposto)	Acompanhar DSA; patch em janela
libc6 (CVEs debsecan)	Alta (base)	Priorizar; pode exigir reboot
Demais CVEs debsecan	Variavel	Triagem semanal; nem todos exigem acao imediata

Discussao: Nem todo CVE debsecan exige patch imediato (falso positivo / mitigacao / ainda sem fix no suite). Criticidade do servico (SSH) eleva prioridade. Atualizar sem planejamento pode causar indisponibilidade — equilibrar via janelas e testes.

```
Terminal - Evidencia 5 - Status de seguranca (equiv. Debian)

root@MachadoPC:~# # Equivalente Debian a ubuntu-security-status
root@MachadoPC:~# dpkg -l | grep -c '^ii'
Pacotes instalados (ii): 547

root@MachadoPC:~# apt-get -s upgrade 2>&1 | grep -iE 'security|upgraded|kept' | head -20
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

# ubuntu-security-status: nao disponivel no Debian
# Equivalente: contagem de pacotes + repositorios security + debsecan
root@MachadoPC:~# grep -r security /etc/apt/sources.list /etc/apt/sources.list.d 2>/dev/null | head -15
/etc/apt/sources.list:deb http://security.debian.org/debian-security trixie-security main

root@MachadoPC:~# apt-cache policy | head -20
Package files:
100 /var/lib/dpkg/status
   release a=now
100 http://ftp.debian.org/debian trixie-backports/main amd64 Packages
   release o=Debian Backports,a=stable-backports,n=trixie-backports,l=Debian Backports,c=main,b=amd64
   origin ftp.debian.org
500 http://security.debian.org/debian-security trixie-security/main amd64 Packages
   release v=13,o=Debian,a=stable-security,n=trixie-security,l=Debian-Security,c=main,b=amd64
   origin security.debian.org
500 http://deb.debian.org/debian trixie-updates/main amd64 Packages
   release v=13-updates,o=Debian,a=stable-updates,n=trixie-updates,l=Debian,c=main,b=amd64
   origin deb.debian.org
500 http://deb.debian.org/debian trixie/main amd64 Packages
   release v=13.6,o=Debian,a=stable,n=trixie,l=Debian,c=main,b=amd64
   origin deb.debian.org
Pinned packages:
```

Figura - Evidencia 5 - Status de seguranca (equiv. Debian)

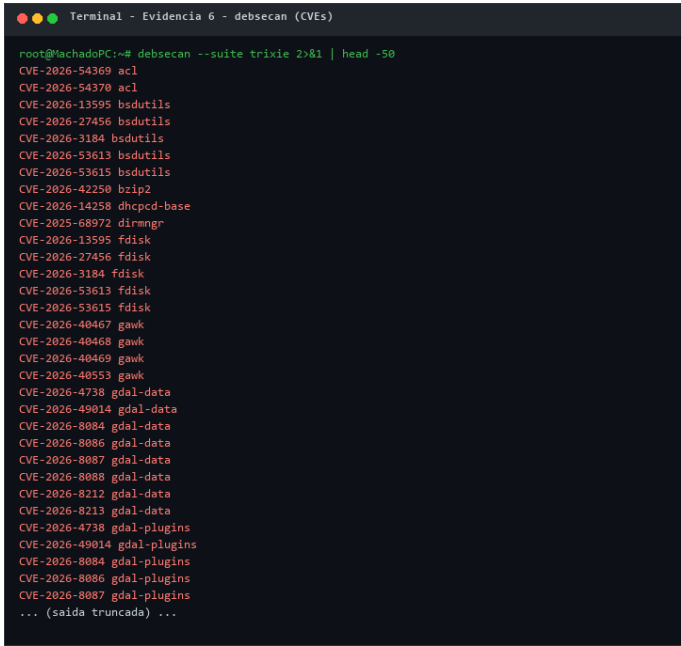


Figura - Evidencia 6 - debsecan (CVEs)

5. Auditoria da Configuracao

Item	Resultado
Lynis	3.1.4 audit system
Hardening Index	69
Warnings	0
Suggestions	48

5 principais recomendacoes:

- 1) Politica de senha (AUTH-9286 / pam_pwquality).
- 2) needrestart apos upgrades (DEB-0831).
- 3) auditd (ACCT-9628).
- 4) Malware scanner periodico (HRDN-7230).
- 5) Fortalecer hashes AIDE SHA256/512 (FINT-4402).

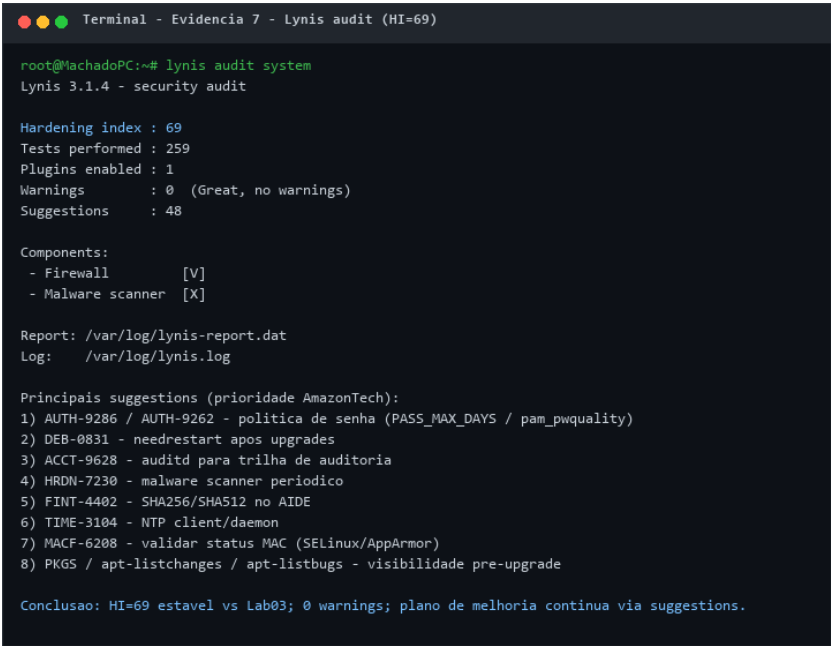


Figura - Evidencia 7 - Lynis audit (HI=69)

6. Verificacao da Integridade

Ferramenta	Resultado
AIDE --check	Diferencas vs baseline (esperado: labels SELinux Lab04A + upgrade)
debsums -c	0 inconsistencias (exit 0)

Alteracoes relevantes? [X] Sim no AIDE — majoritariamente xattrs SELinux e arquivos tocados por manutencao; **nao** tratadas automaticamente como incidente. debsums sem mismatches nos checksums de pacotes. Antes de medida corretiva: classificar mudanca (legitima vs suspeita), correlacionar com change log, so entao atualizar baseline AIDE.

```

Terminal - Evidencia 8 - AIDE --check

root@MachadoPC:~# aide.wrapper --check
AIDE 0.19.1 | baseline: /var/lib/aide/aide.db | exit=7

Start timestamp: 2026-08-06 19:22:26 -0400 (AIDE 0.19.1)
AIDE found differences between database and filesystem!!
Added entries:
Removed entries:
Changed entries:
The attributes of the (uncompressed) database(s):
End timestamp: 2026-08-06 19:23:58 -0400 (run time: 1m 32s)

Entry /auditoria in databases has different attributes: +selinux
Entry /auditoria/parecer.txt in databases has different attributes: +selinux
Entry /bin in databases has different attributes: +selinux
Entry /etc in databases has different attributes: +selinux
Entry /etc/.pwd.lock in databases has different attributes: +selinux
Entry /etc/.updated in databases has different attributes: +selinux
Entry /etc/adduser.conf in databases has different attributes: +selinux
Entry /etc/aide in databases has different attributes: +selinux
Entry /etc/aide/aide.conf in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/01_aide_readcache in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_basenotations in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_bits in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_bootid in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_checksums in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_constants in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_dateformats in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_days in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_distribution in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_hardware in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_hostname in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_logext in databases has different attributes: +selinux
... (saida truncada) ...

```

Figura - Evidencia 8 - AIDE --check

```

Terminal - Evidencia 9 - debsums -c

root@MachadoPC:~# debsums -c 2>&1 | tee /tmp/debsums04b.txt | head -40

exit_code=0 (0=sem inconsistencias reportadas por -c)
linhas alteradas: 0
Nenhuma inconsistencia reportada por debsums -c (checksums OK ou silencio).

root@MachadoPC:~# debsums --help | head -15
debsums checks the MD5 sums of installed debian packages.

Usage: debsums [OPTIONS] [PACKAGE|DEB] ...

Options:
-a, --all                check configuration files (normally excluded)
-e, --config             check only configuration files
-c, --changed            report changed files (implies -s)
-l, --list-missing       list packages which don't have an md5sums file
-s, --silent             only report errors
-m, --md5sums=FILE       read list of deb checksums from FILE
-x, --report-mismatches  report errors and print the md5sums mismatch
-r, --root=DIR           root directory to check (default /)
-d, --admin-dir=DIR      dpkg admin directory (default /var/lib/dpkg)
-p, --deb-path=DIR[:DIR...] search path for debs

```

Figura - Evidencia 9 - debsums -c

7. Consolidacao das Evidencias

Evidencia	Ferramenta	Finalidade
02/03/04 apt	apt	Atualizacao e security patch
05 security status	apt/sources	Cobertura de repositorios security
06 debsecan	debsecan	CVEs conhecidos
08 lynis	Lynis	Auditoria Hardening Index
09 aide	AIDE	Integridade vs baseline
10 debsums	debsums	Checksums de pacotes

8. Plano Permanente de Manutencao do Hardening

Atualizacoes: apt update diario; list --upgradable; security em ate 7 dias (criticos 24-72h); testes em homolog; janela com rollback; registrar evidencias.

Vulnerabilidades: debsecan semanal; acompanhar DSA/security.debian.org; priorizar por exposicao (SSH, libs base); registrar CVE e acao.

Auditorias: Lynis mensal (ou apos mudanca relevante); revisar suggestions; acompanhar evolucao do HI.

Integridade: AIDE check semanal + apos upgrades; debsums mensal; investigar diffs; atualizar DB so apos validacao.

Documentacao: data, responsavel, ferramentas, evidencias, recomendacoes, pendencias.

9. Discussao Tecnica

1. Atualizar aplica patches; manter seguranca continuamente inclui auditoria, integridade, vulnerabilidades e revisao de politica.
2. apt reduz CVEs conhecidos; debsecan prioriza; Lynis acha desvios de config; AIDE/debsums detectam alteracoes — juntos fecham o ciclo.

3. Frequencias:

Frequencia	Atividades
Diariamente	apt update; revisar falhas de auth/Fail2Ban; checar alertas
Semanalmente	apt upgrade security; debsecan; AIDE --check; revisar diffs
Mensalmente	Lynis audit; debsums; revisar plano; atualizar baseline AIDE se OK

- 4. Evidencias indispensaveis:** logs apt, saida debsecan, report Lynis (HI), relatorio AIDE, debsums, registro de mudancas.

- 5. Futuras alteracoes:** change control, re-auditoria Lynis, AIDE check pos-mudanca, menor privilegio e MAC (Lab04A) preservados.

10. Parecer Tecnico

Empresa	AmazonTech
Servidor	WEB-01
Atualizacoes	Em dia apos patch security libexpat1
Vulnerabilidades	Monitoradas via debsecan; triagem continua
Auditoria	Lynis HI=69; 0 warnings
Integridade	AIDE operacional; debsums OK
Data	06/08/2026
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

Riscos: CVEs remanescentes no debsecan (triagem); HI 69 indica melhorias (senha, auditd, malware scanner); diffs AIDE exigem disciplina de baseline.

Recomendacoes prioritarias: (1) ciclo semanal security+debsecan; (2) pam/senha; (3) auditd; (4) atualizar baseline AIDE apos validar Lab04A; (5) needrestart.

Conclusao: [X] **APTO PARA OPERACAO ASSISTIDA** com plano permanente de manutencao instituido. Hardening passa a ser processo continuo ate o Projeto Integrador (Aula 5).

Responsavel: Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

Data: 06/08/2026

11. Conclusoes

Atualizacao, vulnerabilidades, auditoria e integridade sao pilares complementares. Ferramentas usadas (apt, debsecan, Lynis, AIDE, debsums) formam um ciclo operacional reutilizavel. Dificuldade tipica: interpretar volume de CVEs e diffs AIDE sem alarmismo. O Lab 04B prepara a equipe para o Projeto Integrador com checklist e plano permanentes.

Apendice B — Checklist Operacional de Hardening

Atividade	Status	Evidencia
Sistema atualizado	SIM	04_apt_upgrade (libexpat1)
Atualizacoes de seguranca verificadas	SIM	03/04 + security.debian.org
Vulnerabilidades analisadas	SIM	06_debsecan
Auditoria executada	SIM	08_lynis HI=69
Integridade verificada	SIM	09_aide + 10_debsums
Evidencias registradas	SIM	lab04b/evidencias
Plano de manutencao atualizado	SIM	Secao 8 deste relatorio

Screenshots: lab04b/evidencias/shots/img/ | Scripts: lab04b/scripts/